



Genesis Biotech Inc.

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Taq DNA polymerase

Catalog No. PG-30001

Quantity: 500 Unit / 250 ul

Storage: -20°C

Description

Taq DNA polymerase is a recombinant thermostable enzyme modified from *Thermus aquaticus*. The approximate size is 94 kDa. The enzyme catalyzes the polymerization of nucleotides into duplex DNA in the 5'→3' direction in the presence of magnesium and also possesses 5' → 3' exonuclease activity.

Storage buffer

This enzyme is supplied in 20 mM Tris-HCl, pH 8.0, 100 mM NaCl, 0.1 mM EDTA, 0.5 mM DTT, 1% Triton X-100, 50% glycerol.

10X PCR buffer (1.0ml/vial)

The PCR buffer is supplied as a 10X concentrate and should be diluted for use.

Application

Taq DNA polymerase is recommended for use in routine PCR and TA cloning.

Basic PCR Protocol

Add the following components to a sterile 0.5ml microcentrifuge tube sitting on ice:

Components	Volume	Final Concentration
10X PCR buffer (with MgCl ₂)	10 ul	1x
2.5mM dNTP	8 ul	200 uM
forward primer	1-5 ul	0.1-0.5 uM
reverse primer	1-5 ul	0.1-0.5 uM
Taq DNA polymerase(2U/ul)	0.5-2 ul	1-4 unit
Template < 0.5 ug	0.1-0.5 ug for genomic DNA	
	5-50 ng for plasmid	

Sterile distilled water up to 100 ul.

Unit definition

One unit is the amount of enzyme required to catalyze the incorporation of 10 nmol of dNTP into acid-insoluble material in 30 minutes at 74°C, pH9.3, with activated salmon sperm DNA as the template-primer. The reaction conditions are: 25 mM TAPS (pH 9.3 at 25°C), 1 mM 2-mercaptoethanol, 2 mM MgCl₂, 50 mM KCl, 200 uM each of dATP, dTTP, dGTP, 100 uM [α-³²P]dCTP, 0.25 mg/ml activated salmon sperm DNA.

Purity

Nicking activity, endonuclease and exonuclease activity were not detected after the incubation of 0.6 ug of supercoiled pUC18 DNA, 0.6 ug of Lambda DNA or 0.6ug of Lambda-HindIII digest with 10 units of this enzyme for 1 hour at 74°C.

Related products

Hotstart Taq DNA polymerase	PG-30002
Pr Taq DNA polymerase (Taq+Pfu)	PG-30003
Hotstart Pr Taq DNA polymerase (Taq+Pfu)	PG-30004
Klen Taq DNA polymerase	PG-30005
Hotstart Klen Taq DNA polymerase	PG-30006
Anti-Taq Hotstart Antibody	PG-30007
Real-time PCR kit	PG-30009
One Step RT-PCR kit	PG-30010
Genotyping kit	PG-30012
PCR kit	PG-30013

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